Timeline Check in

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Ideas for analysis

Questions

- How does the percent cover of native grasses vary over time compared to the percent cover of exotic vegetation on the NCOS Mesa Perennial Grasslands? - How does intentional disturbance influence this relationship? - How do key individual species, including Stipa pulchra, Medicago polymorpha, Melilotus indicus, and Festuca perennis, vary in percent cover before and after intentional disturbance?

Statistical analysis

- statistical analysis to address at least one question with at least one analysis coded and run

Background

We assess whether the 2023 cultural burn is meeting its restoration goals of promoting native grassland species. This study may help to inform future restoration management decisions regarding intentional disturbance at NCOS and could have implications which extend to the management or ecological understanding of other coastal grassland systems. These questions are relevant because Santa Barbara coastal grasslands are fire adapted (Saglimbeni, 2024, pp. 1-2), but are now heavily affected by invasive species. Studying the NCOS burn helps show whether cultural burning can support native species like Stipa pulchra while reducing nonnative grassland invaders.

Visualizations

Plot 1: Native and exotic percent cover over time

Plot 2: Sweet and bur clover abundance over time

Plot 3: Change in mean percent cover of key species after disturbance

Plot 4: Mean percent cover of native grasses after disturbance

Exploratory Visualization 1: Change in mean percent cover of key species after sheep grazing

Exploratory Visualization 2: Effect of annual precipitation on native and invasive percent cover

Plan for elective

- the general concept
- a description of the progress you have made
- a link, photo, or some visual of your elective